

A note on complex determinism

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Consider a time series,

$$X = \{x_t | t = 1, T\},$$

and its associated segment series in d -dimensional space

$$\begin{aligned} V &= \{x_k | k = t, t-1, \dots, t-d+1\} \\ &\equiv \{v_n(k) | n = 1, \dots, d\}. \end{aligned}$$

Define an indicator function, $I_{ij}(\epsilon)$,¹

$$I_{ij}(\epsilon) = \prod_{k=1}^d \theta(\epsilon - |v_k(i) - v_k(j)|),$$

where θ is the Heaviside step function,

$$\theta(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$$

and

$$C_d(\epsilon) = \frac{1}{M} \sum_{pairs} I_{ij}(\epsilon),$$

where M is the number of pairs of vectors $\{v(i), v(j)\}$, and ϵ is the *resolution*.² $C_d(\epsilon)$ is thus the probability that two d -vectors are within ϵ of each other in all their coordinates.

We now define the quantities, γ_j ,³

$$\gamma_j = 1 - \frac{C_j^2}{C_{j-1}C_{j+1}}.$$

These quantities measure the dependence of $x(i)$ on $x(i-j)$, excluding the trivial induced explicit dependencies for $i-j < k < i$.

¹ P. Grassberger and I. Procaccia, *Measuring the strangeness of strange attractors*, Physica D 9 (1983) 189-208.

² In practice, while ϵ is not defined a priori, it can be defined by a measure of the global width of the time series, assuming it has no trend, such as $1/\sigma$, where σ is the overall standard deviation of the time series.

³ I drop the ϵ for notational simplicity.

As an example, consider a time series formed by a segment of the logistic map, $x(n+1) = \lambda x(n)[1 - x(n)]$, with λ in the chaotic region. Since here $x(j)$ is completely determined by $x(j-1)$, γ_1 will be *large*, and γ_j for $j > 1$ essentially zero; if we intercalate n such maps with different λ s, term by term, only γ_n will be appreciably non-zero. In a noisy version of such a map all γ s can be non-zero, but, still, γ_n will be largest.⁴

⁴ Depending on the amplitude of the noise, of course.